

# Beyond Spatial Accuracy: Diagnosing Persistent Orientation Failures in Vision–Language Models

Anonymous WACV Datasets Track submission

Paper ID \*\*\*\*\*

# Beyond Spatial Accuracy: Diagnosing Persistent Orientation Failures in Vision–Language Models

## Supplementary Material

|     |                                                                                               |     |
|-----|-----------------------------------------------------------------------------------------------|-----|
| 001 | 1. Exact Prompts, Decoding, and Parsing                                                       | 045 |
| 002 | VSR plain prompt (identical for 2B and 7B, all conditions; ‘{statement}’ is the VSR caption): | 046 |
| 003 |                                                                                               | 047 |
| 004 | Look at the image carefully.                                                                  | 048 |
| 005 |                                                                                               | 049 |
| 006 | Statement: ”{statement}”                                                                      | 050 |
| 007 |                                                                                               | 051 |
| 008 | Is this statement true or false?                                                              | 052 |
| 009 |                                                                                               | 053 |
| 010 | Answer with exactly one word:                                                                 | 054 |
| 011 | True or False.                                                                                | 055 |
| 012 | 2B structured-decomposition prompt (2B only; single run):                                     | 056 |
| 013 |                                                                                               | 057 |
| 014 | Look carefully at the image.                                                                  |     |
| 015 |                                                                                               |     |
| 016 | Statement:                                                                                    |     |
| 017 | ”{statement}”                                                                                 |     |
| 018 |                                                                                               |     |
| 019 | Determine whether the statement is true                                                       |     |
| 020 | by doing the following:                                                                       |     |
| 021 | 1. Identify the subject object.                                                               |     |
| 022 | 2. Identify the reference object.                                                             |     |
| 023 | 3. Determine the position and                                                                 |     |
| 024 | orientation of the subject relative                                                           |     |
| 025 | to the reference object.                                                                      |     |
| 026 | 4. Pay careful attention to left/right,                                                       |     |
| 027 | front/back, viewpoint, and                                                                    |     |
| 028 | orientation.                                                                                  |     |
| 029 | 5. Decide whether the stated spatial                                                          |     |
| 030 | relationship matches the image.                                                               |     |
| 031 |                                                                                               |     |
| 032 | Answer with exactly one word:                                                                 |     |
| 033 | True or False.                                                                                |     |
| 034 | SITE multiple-choice prompt (replicated from the                                              |     |
| 035 | official lmms-eval task sitebench/utlis.py):                                                  |     |
| 036 | # image examples (one <image> token per visual):                                              |     |
| 037 | <image>...                                                                                    |     |
| 038 | Question: {question}                                                                          |     |
| 039 | Options:                                                                                      |     |
| 040 | A: {opt0}                                                                                     |     |
| 041 | B: {opt1}                                                                                     |     |
| 042 | ...                                                                                           |     |
| 043 | Give me the answer letter directly.                                                           |     |
| 044 | The best answer is:                                                                           |     |
|     | # video examples (pre-prompt):                                                                | 045 |
|     | Select the best answer to the following                                                       | 046 |
|     | multiple-choice question based on the                                                         | 047 |
|     | video. Respond with only the letter                                                           | 048 |
|     | of the correct option.                                                                        | 049 |
|     | Question: {question}                                                                          | 050 |
|     | Options:                                                                                      | 051 |
|     | A: ...                                                                                        | 052 |
|     | B: ...                                                                                        | 053 |
|     | ...                                                                                           | 054 |
|     | Give me the answer letter directly.                                                           | 055 |
|     | The best answer is:                                                                           | 056 |
|     |                                                                                               | 057 |
|     | Decoding. All generations are greedy:                                                         | 058 |
|     | do_sample=False, temperature 0, top- <i>p</i> 1.0.                                            | 059 |
|     | max_new_tokens: 5 for the 2B wrapper, 128 for                                                 | 060 |
|     | 7B VSR runs, 16 for SITE (change validated parse-                                             | 061 |
|     | identical on 125/125 held-out outputs before adoption;                                        | 062 |
|     | recorded in the SITE run metadata).                                                           | 063 |
|     | Parsing. VSR outputs are parsed with an anchored                                              | 064 |
|     | case-insensitive regex (src/evaluation/parser.py):                                            | 065 |
|     | strips prefixes (‘the answer is’, ‘answer:’, ‘response:’,                                     | 066 |
|     | ‘output:’) and trailing punctuation; accepts exact                                            | 067 |
|     | ‘true/yes/correct’ or ‘false/no/incorrect’ forms; sub-                                        | 068 |
|     | string fallback only when exactly one of ‘true’/‘false’                                       | 069 |
|     | appears. Invalid outputs are counted as wrong, never                                          | 070 |
|     | guessed; invalid rate is 0 on all reported runs. SITE                                         | 071 |
|     | outputs use the official parse_multi_choice_response                                          | 072 |
|     | (letter); unparseable responses are counted incorrect                                         | 073 |
|     | and reported.                                                                                 | 074 |
|     | 2. Training and Adapter Configurations                                                        | 075 |
|     | All LoRA conditions share: $r=8$ , $\alpha=16$ , dropout                                      | 076 |
|     | 0.05, AdamW $\text{lr } 10^{-4}$ , weight decay 0.01, linear                                  | 077 |
|     | warmup 10% of steps, 2 epochs, bf16, gradient                                                 | 078 |
|     | checkpointing, gradient clipping 1.0, vision tower                                            | 079 |
|     | frozen. Adapter configs are committed at check-                                               | 080 |
|     | points/*/final/adapter_config.json.                                                           | 081 |
|     | Manifest construction. General: 2,000 state-                                                  | 082 |
|     | ments sampled from the VSR train split. Tar-                                                  | 083 |
|     | getted: orientation-over-sampled subset of the same                                           | 084 |
|     | split. Hard-negative: the orientation block of the                                            | 085 |
|     | general manifest is replaced by audited-clean originals                                       | 086 |
|     | plus paired hard negatives (facing $\leftrightarrow$ facing away from;                        | 087 |
|     | parallel $\leftrightarrow$ perpendicular) with flipped labels; every in-                      | 088 |
|     | cluded example was audited for label correctness (451                                         | 089 |

| Condition      | LoRA targets          | Trainable | Batch          | Manifest              |
|----------------|-----------------------|-----------|----------------|-----------------------|
| 2B General     | LM q/k/v/o            | ~2.3M     | microbatch 4×4 | general (2,000)       |
| 2B Targeted    | LM q/k/v/o            | ~2.3M     | same           | targeted (2,000)      |
| 7B LM-only     | LM q/k/v/o            | ~0.3M     | 1              | general (2,000)       |
| 7B HardNeg     | LM q/k/v/o            | ~0.3M     | 1              | hard-negative (2,000) |
| 7B Projector   | visual.merger.mlp.0/2 | 152K      | 1              | general (2,000)       |
| 7B Vision+Proj | merger + blocks 24–31 | 1.46M     | 1              | general (2,000)       |

Table 1. LoRA conditions and their target modules.

train orientation examples audited: 402 clean, 28 excluded, 21 original-only). 2B runs split 5% of the manifest for validation (seed 42). All manifests derive exclusively from the VSR train split; all evaluations use test only.

### 3. Probe Methodology and Full Tables

Features. Qwen2-VL-7B-Instruct, vision tower frozen. ViT patch embeddings (1280-d) and post-merger features (3584-d), mean-pooled per image (ungrounded) or per grounding box (grounded). Grounding boxes from frozen Qwen2-VL grounding (94.6% of 647 orientation images yield both boxes; failures are mostly genuinely absent objects), rescaled to pixels; region features mean-pooled over patch centers inside each box.

Tasks. T1 facing vs. facing away (test  $n=103$ , majority 64.2%); T2 parallel vs. perpendicular ( $n=34$ , majority 64.7%); T3 4-way ( $n=137$ , majority 49.6%). Training on audited-clean VSR train; 5-fold CV; no test examples in training.

Readouts. Linear logistic regression; MLP (256 hidden, ReLU, Adam, early stopping on CV); patch-vote (per-patch linear probe, majority vote over patches); grounded variants with feature sets visual ([subj, ref, subj-ref, subj-ref]), geometry (12-d normalized box features), and visual+geometry.

Ungrounded results (test acc %; balanced acc in parentheses; Wilson 95% CI; majority baselines: T1 64.2%, T2 64.7%, T3 49.6%):

| T1 facing vs. away ( $n=103$ ) | Test | Bal. acc | 95% CI   |
|--------------------------------|------|----------|----------|
| ViT linear                     | 65.0 | 60.4     | [55, 74] |
| ViT MLP                        | 55.3 | 52.0     | [45, 65] |
| Merger linear                  | 68.9 | 64.5     | [59, 77] |
| Merger MLP                     | 52.4 | 49.2     | [42, 62] |
| ViT patch-vote                 | 62.1 | —        | [53, 71] |
| Merger patch-vote              | 60.2 | —        | [51, 69] |

| T2 parallel vs. perpendicular ( $n=34$ ) | Test | Bal. acc | 95% CI   |
|------------------------------------------|------|----------|----------|
| ViT linear                               | 61.8 | 61.0     | [44, 77] |
| ViT MLP                                  | 67.6 | 69.3     | [50, 82] |
| Merger linear                            | 64.7 | 61.4     | [47, 79] |
| Merger MLP                               | 64.7 | 63.3     | [47, 79] |
| ViT patch-vote                           | 73.5 | —        | [57, 85] |
| Merger patch-vote                        | 67.6 | —        | [51, 81] |

| T3 4-way ( $n=137$ ) | Test | Bal. acc | 95% CI   |
|----------------------|------|----------|----------|
| ViT linear           | 52.6 | 40.5     | [44, 61] |
| ViT MLP              | 52.6 | 42.9     | [44, 61] |
| Merger linear        | 53.3 | 44.0     | [45, 61] |
| Merger MLP           | 47.4 | 43.2     | [39, 56] |
| ViT patch-vote       | 51.8 | —        | [44, 60] |
| Merger patch-vote    | 52.6 | —        | [44, 61] |

Grounded results (CV acc / test acc; majority T1 63.7%, T2 57.0%; visual = [subj, ref, subj-ref, subj-ref]; geometry = 12-d box features; T3 4-way is at chance for every readout, test acc 41.7–52.8% across all 12 cells vs. 49.9% majority, committed JSON):

| Readout       | T1 visual | T1 geom.  | T2 visual | T2 geom.  |
|---------------|-----------|-----------|-----------|-----------|
| ViT linear    | 59.8/61.6 | 61.1/60.6 | 51.2/53.6 | 44.2/50.0 |
| ViT MLP       | 59.5/66.7 | 61.8/62.6 | 45.4/46.4 | 45.4/46.4 |
| Merger linear | 57.2/65.7 | 61.1/60.6 | 51.2/53.6 | 44.2/50.0 |
| Merger MLP    | 56.3/52.5 | 61.1/71.7 | 49.0/53.6 | 50.0/42.9 |

The single above-majority cell (geometry-only merger MLP on T1, 71.7% test) has CV 61.1% below the 63.7% majority — not robust — and derives from the grounding model’s own box statistics, not vision-feature content. In short, object-box conditioning with the model’s own grounding predictions and mean-pooled region features does not recover robust orientation decodability under the tested readouts. The T2 patch-vote result (73.5%) is a suggestive point estimate with a wide CI overlapping the majority baseline ( $n=34$ ).

Clear-subset control (canonical frozen snapshot). The 48-example manual audit identified ambiguous or questionable orientation labels. Re-evaluation on progressively stricter subsets (values % from results/tables/orientation\_clean\_label\_table.md, generated by scripts/clean\_label\_orientation.py): Because auditing changes the evaluated sample, strict-subset values are not a matched comparison against other relation families. The earlier probe-side clear/strict check (pre-freeze) additionally suggested that removing audit-flagged examples does not change the probe conclusion: the best ungrounded probe moved only 68.9% → 71.1% → 71.6%

Table 2. Clean-label orientation robustness (canonical frozen snapshot; values % from results/tables/orientation\_clean\_label\_table.md).

| Condition                | Full (137) | −Questionable (132) | Clear (124) | Strict (107) |
|--------------------------|------------|---------------------|-------------|--------------|
| 2B zero-shot             | 62.8       | 65.2                | 66.9        | 75.7         |
| 2B structured            | 53.3       | 52.3                | 51.6        | 53.3         |
| 2B General LoRA          | 62.0       | 63.6                | 64.5        | 72.9         |
| 2B Targeted LoRA         | 64.2       | 65.9                | 68.5        | 76.6         |
| 7B zero-shot             | 63.5       | 65.2                | 68.5        | 72.0         |
| 7B General LoRA          | 65.7       | 67.4                | 70.2        | 78.5         |
| 7B Targeted LoRA         | 64.2       | 65.9                | 69.4        | 72.0         |
| 7B Hard-Neg LoRA         | 66.4       | 68.2                | 70.2        | 76.6         |
| 7B Projector LoRA        | 64.2       | 65.9                | 69.4        | 73.8         |
| 7B Vision+Projector LoRA | 64.2       | 65.2                | 68.5        | 73.8         |

(full/clear/strict,  $n = 103/97/88$ ) and the generative T1 readouts 64.1/66.0/68.2 (zero-shot) and 68.9/72.2/78.4 (General LoRA); committed in results/probe/clear\_subset\_results.json.

#### 4. Complete Statistical Test Battery

All pairwise tests are exact McNemar tests (binomial test on discordant pairs, two-sided). Discordant counts are reported as (condition-loses / condition-gains) relative to the control named in each row. No test was run and withheld; the battery below is complete for this paper.

Principal vs. exploratory comparisons. The battery above is fully enumerated; no test was run and withheld. The principal (confirmatory) comparisons are: zero-shot vs. LM-only consistency (ID 19) and the SITE transfer tests (§E). All other comparisons — vision-side conditions and their per-relation cells, hard-negative training vs. control, the remaining consistency comparisons, and the two-stage agreement rows (IDs 15–18) — are exploratory diagnostics reported with exact unadjusted  $p$ -values. Under a Holm adjustment over the 18 accuracy-valid VSR tests ( $\alpha=0.05$ ), only the zero-shot-vs-LM-only consistency test survives (Holm  $p=0.0018$ , using the conservative  $p<0.0001$  bound); the vision-side overall effects (raw  $p=0.0043$ ,  $p=0.0122$ ; Holm 0.073, 0.195) and all other comparisons do not survive a blanket adjustment and are not load-bearing. For the 3 SITE transfer tests, Holm-adjusted  $p$ -values are 0.012 (primary), 1.0, 1.0.

Single-checkpoint note. Every trained condition is a single checkpoint. All McNemar tests above quantify paired disagreement on the fixed test examples between two conditions; they do not estimate training-run variability, and the single-seed results reported here do not estimate training-run variability. Adapter-rank claims (e.g., the 1–2pp overall differences among trained 7B conditions) should be read in that light.

Consistency verdict patterns. For each strict

family and condition, the committed verdict table (results/consistency\_verdict\_table.json, scripts/consistency\_verdict\_table.py) reports the observed both-True / both-False / complementary counts, the original and flipped True rates, and the verdict rates expected under verdict independence given those base rates:

The zero-shot pattern is dominated by a False-defaulting response bias: expected consistency under verdict independence is 30.2%, below the observed 36.9%, so a simple base-rate null does not explain the facing inconsistency; conversely, training raises consistency while introducing both-True contradictions (0%  $\rightarrow$  24.3% LM-only / 17.5% hard-neg), a disclosed specialization trade-off. Full table for left/right ( $n=245$ ), front/behind ( $n=314$ ), and parallel/perp (soft,  $n=34$ ) in the committed JSON.

Consistency definitions. Strict complements (exactly one truth value): left $\leftrightarrow$ right ( $n=245$ ), in front of $\leftrightarrow$ behind ( $n=314$ ), facing $\leftrightarrow$ facing away ( $n=103$ ). Soft complement (parallel $\leftrightarrow$ perpendicular,  $n=34$ ): both-False is legitimate; only both-True is a contradiction; both-True rates are reported separately. Self-consistency on a strict pair means opposite verdicts.

Wilson CIs. All per-relation and SITE subset accuracies are reported with Wilson 95% confidence intervals (main text and committed JSONs).

#### 5. SITE Protocol, Keyword Rule, and Distributions

Pre-specification. Protocol frozen 2026-08-09 before any evaluation; example IDs committed in results/site/site\_protocol.json; the protocol configuration is recorded in results/site/run\_metadata.json. Evaluation-only: no SITE data used for training. Evaluation: Qwen2-VL-7B-Instruct frozen, greedy, official multiple-choice prompt and letter parser, max\_new\_tokens=16, images resized to  $\leq 392$  px long side (constant protocol parameter; the installed trans-

| ID   | Comparison                 | Subset                | Discord. | $p$       |
|------|----------------------------|-----------------------|----------|-----------|
| 1    | HardNeg vs. Gen.           | overall (2195)        | 52/60    | 0.508     |
| 2    | HardNeg vs. Gen.           | weak families         | 20/26    | 0.461     |
| 3    | Projector vs. ctrl.        | overall               | —        | 0.0043    |
| 4    | V+Proj vs. ctrl.           | overall               | —        | 0.0122    |
| 5    | Projector vs. ctrl.        | orientation           | —        | 0.86      |
| 6    | V+Proj vs. ctrl.           | orientation           | —        | 0.85      |
| 7–14 | Proj./V+Proj vs. ctrl.     | per-relation (4×2)    | —        | 0.25–1.00 |
| 15   | Two-stage A-lin. vs. ctrl. | agreement (127 boxed) | 45/16    | 0.0003    |
| 16   | Two-stage A-MLP vs. ctrl.  | agreement (127 boxed) | 47/14    | <0.0001   |
| 17   | Two-stage B-lin. vs. ctrl. | agreement (127 boxed) | 44/20    | 0.0037    |
| 18   | Two-stage B-MLP vs. ctrl.  | agreement (127 boxed) | 47/18    | 0.0004    |
| 19   | Zero-shot vs. LM-only      | consistency 662       | 117/43   | <0.0001   |
| 20   | HardNeg vs. LM-only        | consistency 662       | 31/41    | 0.29      |
| 21   | Projector vs. LM-only      | consistency 662       | 63/48    | 0.18      |
| 22   | V+Proj vs. LM-only         | consistency 662       | 65/56    | 0.47      |

Table 3. Complete VSR test battery (22 rows). Per-relation cells (IDs 7–14) are enumerated in the committed vision\_side\_ comparison.json: projector facing  $p=0.77$ , facing-away  $p=1.00$ , parallel  $p=0.25$ , perpendicular  $p=1.00$ ; vision+proj facing  $p=1.00$ , facing-away  $p=0.58$ , parallel  $p=1.00$ , perpendicular  $p=1.00$ . Rows 15–18 are agreement-based: the committed two-stage pipeline scored agreement with the control’s verdicts (not accuracy against ground truth), so these rows are descriptive and are excluded from the confirmatory set and from the Holm family. SITE transfer tests (3: all/primary/secondary, §E) form a separate small family.

Table 4. Facing-pair verdict patterns by condition ( $n=103$ ); “exp.” = expected under verdict independence given the observed base rates.

| Condition (facing pairs, $n=103$ ) | both-T     | both-F     | comp.      | orig-T | flip-T | exp. both-F | exp. comp. |
|------------------------------------|------------|------------|------------|--------|--------|-------------|------------|
| 7B zero-shot                       | 0          | 65 (63.1%) | 38 (36.9%) | 0.20   | 0.17   | 66.5%       | 30.2%      |
| LM-only LoRA                       | 25 (24.3%) | 10 (9.7%)  | 68 (66.0%) | 0.62   | 0.52   | 18.0%       | 49.4%      |
| HardNeg LoRA                       | 18 (17.5%) | 5 (4.9%)   | 80 (77.7%) | 0.60   | 0.52   | 19.0%       | 50.2%      |
| Projector LoRA                     | 18 (17.5%) | 15 (14.6%) | 70 (68.0%) | 0.53   | 0.50   | 23.5%       | 50.3%      |
| Vision+Proj LoRA                   | 14 (13.6%) | 23 (22.3%) | 66 (64.1%) | 0.49   | 0.43   | 27.1%       | 49.4%      |

formers build ignores max\_pixels), SDPA attention.

Subsets. Primary = official category spatial relationship reasoning (1,721 examples; 993 images). Secondary = heuristic orientation keywords (3,272; 2,024 images). Exploratory = official movement prediction & navigation (1,103; 248 images). Overlaps (image): primary  $\cap$  secondary = 488, secondary  $\cap$  exploratory = 186, primary  $\cap$  exploratory = 0; union = 2,591 images, each evaluated once.

Exact orientation-keyword rule (applied, case-insensitive, word- boundary regex, to question text plus non-image option text; src/datasets/site.py):

orientation, oriented, orient,  
facing, face, faces, direction,  
directional, view, viewpoint,  
view angle, angle, turned, rotate,  
rotated, rotation, toward, towards,  
away from, left, right, front,  
behind, in front of, parallel,  
perpendicular, clockwise,  
counterclockwise

Secondary subset composition ( $n=2,024$  images) by

official category: multi-view & cross-image reasoning 866 (raw 37.8%, CAA 8.8%); spatial relationship reasoning 488 (71.3%, 53.5%); object localization & positioning 346 (46.5%, 28.3%); movement prediction & navigation 186 (49.5%, 24.3%); 3D information understanding 91 (27.5%, 0.4%); counting & existence 47 (53.2%, 37.1%). Top sources: exoego4d 316 (36.4%), egoexo4d 288 (34.7%), MMTBench 237 (48.9%), SAT 183 (46.4%), MMIU 133 (28.6%), CLEVR 106 (92.5%), SPEC 102 (50.0%), SpatialEval 100 (53.0%), Three-DSRBench 95 (61.1%), MMERealWorld 89 (32.6%), SeedBench 63 (58.7%), GQA 41 (75.6%), VStarBench 37, Blink 35, plus 27 smaller sources.

Decision rule (pre-specified). (1) If the primary subset shows the same broad weakness: run the VSR-trained 7B General LoRA on SITE (pre-specified step 2). (2) Else if only the orientation subset shows the effect: narrow the claim to object/direction orientation. (3) Else: report benchmark-specificity. Outcome: branch 2; the claim was narrowed accordingly. The VSR-trained 7B General LoRA adapter was subsequently evaluated on the iden-

tical SITE examples as a post-protocol cross-dataset transfer check (no SITE training; predictions in results/site/vsr\_lora\_predictions.csv, paired statistics from scripts/compare\_site\_zeroshot\_lora.py, canonical table in results/tables/site\_validation\_table.md):

Entries are raw accuracy / CAA (%). The adapter is statistically neutral overall, does not improve the orientation subset, and significantly degrades the official spatial-relationship category — VSR gains transfer poorly and can induce negative transfer.

Composition-confound analysis (CPU-only, zero-shot predictions). Because the orientation slice is keyword-derived and heterogeneous, we test whether the heuristic flag predicts lower correctness after controlling for task composition (logistic regression over the 2,591 zero-shot predictions; scripts/site\_confound\_analysis.py → results/site/orientation\_confound\_analysis.json):

Within the official spatial relationship reasoning category, orientation-flagged questions remain descriptively 7.5pp harder than unflagged ones (71.3% vs. 78.8%;  $n=488/505$ ). No source dataset has  $n \geq 30$  in both arms (orientation-heavy corpora are almost entirely flagged, others almost never), so source indicators absorb most of the orientation variance in the full model. Reading: the adjusted association is in the expected direction but not significant; the low aggregate heuristic score is largely explained by task composition. We therefore do not treat SITE as independent confirmation of the VSR orientation bottleneck, and the main text reports the slice as exploratory.

High-precision post-hoc subset (exploratory only). Restricting the heuristic to VSR-construct terms only (facing, facing away, parallel, perpendicular; word-boundary match on question or option text) yields  $n=177$  examples (77 in the official spatial-relationship category, 99 in multi-view/cross-image reasoning, 1 in object localization): raw accuracy 44.1% (95% CI [37.0, 51.4]), CAA -2.3%. This subset is explicitly post-hoc and does not replace the pre-specified frozen definition; it is reported as exploratory only.

## 6. Failure Taxonomy

The 48 persistent-failure statements are the union of: wrong in both 7B zero-shot and 7B LoRA (20); wrong in  $\geq 3$  of 4 base conditions (38); wrong in both 2B and 7B zero-shot (28). A single-annotator exploratory audit (eight-class taxonomy) suggests many persistent errors remain visually interpretable while also identifying viewpoint and annotation ambiguities; it is qualitative evidence, not a causal localization of the failure. Annotations committed in results/failure\_annotations.csv and results/orientation\_persistent\_annotations.csv.

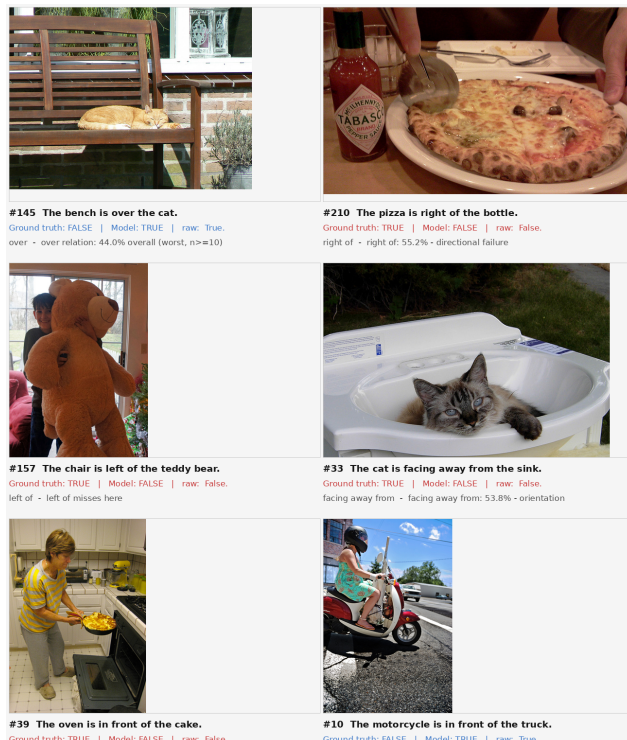


Figure 1. Representative persistent failures (annotated grid; 15 cases). Images: VSR test set. Per the single-annotator exploratory audit, the most frequent class is clear-image reasoning failure: the pose is visually interpretable to a human and the model still errs.

Transition counts. Scaling (2B→7B zero-shot): 28 both wrong, 23 2B-wrong/7B-right, 22 2B-right/7B-wrong, 64 both right. Adaptation (7B zero→7B LoRA): 20 both wrong, 30 fixed, 27 broken, 60 both right. Of the 48 annotated cases, hard-negative training fixes 5; 15 still fail under all three 7B conditions. Facing/facing-away account for 68.8% of persistent failures.

Inter-annotator agreement (additive reliability evidence). Because the clean-label audit (§3) and the taxonomy above are single-annotator, the accompanying code package includes a blind second-annotation protocol: scripts/export\_iaa\_sheets.py exports two rating sheets over the same examples (the 137-example orientation test set for the binary clean/ambiguous flag, and the 48 persistent-failure cases for the eight-class taxonomy), containing only image, statement, and relation—ground truth, model predictions, and the first annotator’s labels are withheld—and scripts/iaa\_tool.py provides a blind annotation interface. A second, independent human annotator rates the same examples under this protocol; agreement is then computed by scripts/compute\_iaa.py as Cohen’s kappa for the bi-

Table 5. SITE VSR-LoRA transfer check (post-protocol; predictions in results/site/vsr\_lora\_predictions.csv, canonical table in results/tables/site\_validation\_table.md). Subsets are defined by the frozen protocol IDs (see note below); no keyword reconstruction is used in analysis.

| Subset                               | Zero-shot   | VSR-LoRA    | $\Delta$ | McNemar $p$ |
|--------------------------------------|-------------|-------------|----------|-------------|
| All images ( $n=2,591$ )             | 54.2 / 31.1 | 53.8 / 30.5 | -0.4     | 0.66        |
| Primary: spatial-rel. ( $n=993$ )    | 75.1 / 59.2 | 71.6 / 53.4 | -3.5     | 0.004       |
| Secondary: orient. kw. ( $n=2,024$ ) | 48.3 / 24.0 | 48.7 / 24.5 | +0.4     | 0.70        |
| Exploratory: movement ( $n=248$ )    | 48.8 / 23.1 | 50.4 / 25.6 | +1.6     | 0.64        |

Table 6. Composition-confound logistic regressions (2,591 zero-shot SITE predictions; outcome = correct).

| Model: correct $\sim$ orient + controls     | OR (orient) | 95% CI       | $p$                   |
|---------------------------------------------|-------------|--------------|-----------------------|
| Unadjusted                                  | 0.31        | [0.25, 0.38] | $2.4 \times 10^{-28}$ |
| + official category                         | 0.74        | [0.57, 0.96] | 0.023                 |
| + category, source, modality, $n_{options}$ | 0.84        | [0.60, 1.16] | 0.29                  |

| Failure mode                          | n  | Share |
|---------------------------------------|----|-------|
| Clear image, model reasoning failure  | 18 | 37.5% |
| Camera viewpoint ambiguity            | 8  | 16.7% |
| Parallel/perpendicular geometry       | 6  | 12.5% |
| Annotation questionable (label wrong) | 5  | 10.4% |
| Intrinsic orientation ambiguous       | 4  | 8.3%  |
| Front/back object ambiguous           | 4  | 8.3%  |
| Small / occluded object               | 2  | 4.2%  |
| Subject/reference inversion           | 1  | 2.1%  |

Table 7. Persistent-failure taxonomy, 48 annotated cases.

nary flag and Krippendorff’s alpha (nominal) for the taxonomy, both with bootstrap 95% confidence intervals, and written to results/iaa\_results.json with a summary at results/iaa\_summary.md. If the second rater is not yet available, these statistics are simply absent: the audit remains single-annotator, and no reported accuracy number (Table 2) is altered by this protocol. An LLM-assisted relabeling, if ever used, would be reported separately as a weaker, non-human-IAA check and would not be substituted for the human second rater.

## 7. Reproducibility and Artifact Index

Code: anonymized archive accompanying this submission (no external links). Every headline number in this paper maps to a committed artifact; the audit script scripts/make\_paper\_figures.py recomputes the main tables directly from the raw prediction CSVs (output archived at paper/fig/audit\_output.txt).

Environments. VSR/7B and SITE runs: Ubuntu 24.04, Python 3.12.13, torch 2.12.1+cu130, transformers 5.14.1, datasets 5.0.1, peft 0.20.0, 1× RTX A6000 (48 GB). 2B runs: earlier python3.10-era environment (SmolVLM2 wrapper). No lock files exist; the adapter

configs (committed) pin the LoRA hyperparameters. Seeds: 42 defaults in 2B paths; single default seed elsewhere — per-run seeds are not recorded in the metrics JSONs (a known limitation).

Canonical freeze. The experimental result set is a frozen snapshot. Canonical tables are generated by scripts/build\_canonical\_tables.py; clean-label results by scripts/clean\_label\_orientation.py; publication figures by scripts/make\_publication\_figures.py; SITE paired transfer statistics by scripts/compare\_site\_zeroshot\_lora.py; VSR/SITE headline recomputation by scripts/make\_paper\_figures.py (output archived at paper/fig/audit\_output.txt).

Two-stage reproducibility caveat. The two-stage pipeline requires per-patch embeddings (results/probe/patch\_embeddings.pkl, 8.2 GB, not committed; regenerable on GPU via scripts/extract\_patch\_embeddings.py) and hard-codes a Linux working directory. The committed evidence is the aggregate results/two\_stage\_results.json (agreement-based, see App. D); the per-example decisions are not committed.

SITE secondary-subset audit note (corrected before submission). An earlier derived analysis (compare\_site\_zeroshot\_lora.py v1, which fed the initial zeroshot\_image\_metrics.json-adjacent figures and the first canonical site table) reconstructed the secondary subset by applying ORIENTATION\_KEYWORDS to the question text only, yielding  $n=1,824$  (47.3% raw / 22.6% CAA). The frozen protocol’s pre-specified definition — results/site/site\_protocol.json  $\rightarrow$  frozen\_ids.secondary (3,272 examples; 2,024 with images), intersected with the 2,591 evaluated IDs — is the authoritative subset, and all numbers in this paper use it ( $n=2,024$ ;

| Headline number                                | Artifact                         | Commit |
|------------------------------------------------|----------------------------------|--------|
| 74.0 / 76.6 / 76.5 / 68.3 (2B)                 | smolvlm2_*_metrics_*.json        | —      |
| 80.9 / 84.7 / 84.3 / 82.9 / 83.1 (7B)          | *_metrics_*.json                 | —      |
| 62.8 / 63.5 / 65.7 / 66.4 (orient. family)     | same + prediction CSVs           | same   |
| Per-relation cells (Table 1 (main))            | orientation_analysis.json        | —      |
| 48-failure taxonomy                            | orientation_persistent_*.json    | —      |
| Probe tables (App. 3)                          | probe/*.json                     | —      |
| Two-stage (App. D, agreement-based rows 15–18) | two_stage_results.json           | —      |
| Consistency (Fig. 4 (main))                    | consistency_stats_all.json       | —      |
| Vision-side p-values                           | vision_side_comparison.json      | —      |
| SITE numbers                                   | site/zeroshot_7b_predictions.csv | —      |
|                                                | site/zeroshot_image_metrics.json | —      |

Table 8. Headline-number → artifact → commit index (all paths relative to results/).

407 zero-shot 48.3% / 24.0% CAA; VSR-LoRA 48.7% /  
 408 24.5%,  $p=0.70$ ). The derived question-only figure was  
 409 corrected before submission; both definitions support  
 410 the same conclusion (orientation  $\ll$  primary), but only  
 411 the frozen-ID set matches the pre-specification. No  
 412 prediction CSV was modified and no model was rerun.  
 413 Data licenses/access. VSR: cam-  
 414 bridgelt/vsr\_random (HF). SITE: franky-  
 415 veteran/SITE-Bench, CC-BY-4.0 (media in 130 GB  
 416 zips; requires HF read token). LoRA adapters are  
 417 committed and loadable with peft; base-model weights  
 418 via HF Hub.  
 419 Exact artifacts for the probe tables  
 420 (App. 3): results/probe/probe\_results.json,  
 421 results/probe/patch\_probe\_results.json, re-  
 422 sults/probe/grounded\_probe\_results.json, re-  
 423 sults/probe/clear\_subset\_results.json, re-  
 424 sults/probe/grounded\_boxes.json.